

Using Segment Anything (SAM) to segment cancer cells in H&E Images

Yanis Chamson
CentraleSupélec

yanis.chamson@student-cs.fr

Olav De Clerck
CentraleSupélec

olav.declerck@student-cs.fr

Deepti Sammeta
CentraleSupélec

deepti.sammata@student-cs.fr

Supervisors: Maria Vakalopoulou
Stergios Christodoulidis

Abstract

Cancer is ranked as the second leading cause of death worldwide. Its rapid detection by physicians is therefore of paramount importance. However, this process is currently time-consuming and can lead to small metastases being missed. Technological advancements such as the digitization of glass slides in pathological laboratories and the development of artificial intelligence (AI) algorithms allow for the detection of cancer cells in hematoxylin and eosin-stained (H&E) tissue images. Numerous automatic detection algorithms have already been proposed and demonstrate good performance. However, the recent emergence of the Segment Anything Model (SAM), which is a foundational model for segmentation, could also play a role in more accurate detection of cancer cells. This project studies how SAM, a segmentation model trained for natural image segmentation (i.e., non-medical), could be used in the context of detecting cancerous cells in H&E images. We thus demonstrate that employing SAM as a post-processing step to HoverNet increases the segmentation quality by 0.18 dice score and by 277.30 in terms of Hausdorff distance. Additionally, we demonstrate that using MedSAM an iteration of SAM pretrained on medical images provides a slight improvement of 0.01 on Dice score comparing to cellViT state of the art. The code is publicly available at https://github.com/yanis92300/Lab_project.git

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1. Introduction

Cancer is a severe disease burden worldwide, with millions of new cases yearly and ranking as the second leading cause of death after cardiovascular diseases [25]. The surgical procedure consists of dissecting the tumor's microenvironment in order to identify and examine it. The tissues are then stained to obtain hematoxylin and

eosin-stained (H&E) images and examined by pathologists in order to identify cancerous regions. This tedious examination process is nevertheless time-consuming and suffers from a high intra- and inter-observer variability.

The recent development of technologies enabling the digitization of H&E images opens the door to the application of computer vision (CV) techniques that can aid in the development of a cheaper, faster, and, most importantly, more precise detection process. Techniques for analyzing H&E images often rely on CNNs [2] [17] [12] and offer clinical-grade results [2]. More recently, however, we have witnessed the development of transformers [26], and their vision transformer (ViT) variant in CV [6], which has led to a significant leap in performance in image processing. These technologies have also been leveraged in the context of automatic metastasis detection and currently constitute the SOTA. [11] [24]

Segment Anything (SAM) [15] is a ViT-based model architecture equipped with a heavyweight image encoder and a lightweight mask decoder. Trained on over 1 billion masks from 11 million images, SAM exhibits excellent segmentation capabilities on various "natural" (i.e. non medical) images. However, its performance on microscopic images, particularly H&E images, falls short of expectations, due to its lack of medical-specific knowledge, including challenges like low image contrast, ambiguous tissue boundaries, and tiny lesion regions [13] [20] [28] [18].

2. Research Objectives

The research question addressed in this project is to investigate how SAM can be leveraged for automatic cancer cell detection in H&E images. Specifically, the objectives are:

1. Combine elements from the literature using SAM to

enhance the performance of the current state-of-the-art (SOTA) in nuclei segmentation (i.e. CellViT [11]).

2. Assessing SAM’s effectiveness as a post-processing tool for classic segmentation models.

3. Dataset

For our study, we have used the PanNuke dataset [7], which is renowned for its diverse collection of annotated histopathological images. This dataset extensively covers a range of tissue types and provides annotations for various nuclei classes, including inflammatory, neoplastic, connective, dead, and non-neoplastic epithelial cells. The PanNuke dataset is particularly notable for representing 19 different types of cancer, which adds significant value to its comprehensiveness and applicability in medical image analysis, especially in cancer research.

However, it is important to note that the dataset exhibits a degree of class imbalance, a common challenge in medical image datasets. For instance, the number of annotations for neoplastic cells is significantly higher, with 77,403 instances, compared to dead cells, which are represented with only 2,908 annotations. Inflammatory and connective cells are presented with 32,276 and 50,585 instances, respectively, while non-neoplastic epithelial cells have 26,572 instances. This imbalance reflects the varying prevalence of cell types in real-world conditions and highlights the importance of designing our model to effectively learn from both abundant and rare data points.

4. Related work

Approaches to automatic detection of cancerous cells can typically take two forms: (i) classification and (ii) segmentation. In both cases, the deep learning (DL) algorithm cannot directly process the whole slide image (WSI) images as they are in the form of very high-resolution images that are very memory-intensive, up to 4 GB. Hence, these images are typically divided into image patches that are processed individually.

In the context of classification, we find ourselves in a situation of weakly-supervised learning and, more specifically, Multiple-Instance-Learning (MIL). In such a setting, we thus have a label for the (WSI) that concerns a multitude of patches which are treated individually. One of the main techniques in the MIL setting is clustering-constrained attention multiple instance learning (CLAM), which uses attention-based learning to automatically identify sub-regions of high diagnostic value in order to accurately classify the WSI. The strongly attended and weakly attended regions are used as representative samples to train clustering layers that learn a rich patch-level

feature space separable between the positive and negative evidence of distinct classes [17]. Another major actor in the MIL landscape is TransMIL which addresses the neglected correlation between different instances in MIL by exploring the morphological and spatial information between different instances [24].

We nevertheless decided to focus on segmentation given that it is the most popular approach in the literature and that we started our project shortly after the release of SAM which caused a wave of enthusiasm in the literature of AI in the medical field. [30]

The weak supervision context does not affect the segmentation because each patch can simply be segmented individually. The most classic techniques are based on non-DL algorithms such as the work of or instance, Cheng and Rajapakse [5], Veta et al. [27], and Ali and Madabhushi [1] that rely on a predefined nuclei geometry and the watershed algorithm to separate clustered nuclei. The major disadvantage is their dependence on handcrafted features and their sensitivity to hyperparameters. DL makes it possible to automate this extraction of features while having much fewer parameters. The invention of the U-net by Ronneberger et al. has become the baseline regarding segmentation [22]. It is a fully convolutional architecture of a u-shape encoder-decoder with skip connections at multiple network depths which offers good results especially for segmentation tasks. Since its introduction, the model has been improved to achieve self-configuration of its parameters [14] or to add attention gates that filter the features propagated through the skip connections, helping the model focus on relevant features while suppressing irrelevant or noisy information. [19]

HoverNet, another SOTA approach for automatically segmenting nuclei instances involves utilizing the horizontal and vertical distances between nuclei pixels and their center of mass. The method separates the nuclei by employing the gradient of the horizontal and vertical distance maps as an input for an edge detection filter (Sobel Operator). [8]

There have also been initiatives aimed at introducing self-supervision techniques where models are trained to predict the magnitude order of an image. It has been shown that identifying the magnification level for tiles can generate a preliminary self-supervision signal to locate nuclei, and it is possible to retrieve meaningful segmentation maps as an auxiliary output to the primary magnification identification task. [23]

Finally, in recent years, different uses of ViT for seg-

mentation have been presented [3] [16] [10] [9] [29] [31]. The first initiatives were to incorporate Transformers into U-Net shaped architectures, as Attention Unet [19] did, or as TransUnet [3] which uses a transformer to encode tokenized patches from a CNN feature map as the input sequence to derive global context within the CNN network. The Segformer model by Xie et al [29] incorporates an adapted Transformer as an image encoder connected to a lightweight MLP decoder segmentation head. In contrast to these methods, the SETR model [31], uses the original ViT as encoder and a fully convolutional network as decoder, both connected without intermediate skip connections.

CellViT, the current best performing model in terms of nuclei-segmentation, combines the SETR model with the U-net and the Hovernet. It thus features a ViT-style encoder with a Hovernet-style decoder, all with skip connections inspired by U-net [11]. Given the limited amount of available data in the medical domain, pre-trained models are an essential requirement as ViTs have increased data requirements compared to CNNs. The authors thus tried leveraging the weights of two different ViTs. The first are the weights of a pre-trained ViT on 104 million histological images [4]. The other is the heavy-weight encoder from SAM [15], which actually holds all the power of SAM as it is trained on over 1 billion masks. The authors experimentally observed that the best segmentation results are obtained when the encoder used is that of SAM.

Another model aimed at leveraging SAM for medical segmentation is MedSAM [18]. It extends from the SAM model, but is further trained on a dataset of 1,090,486 medical image-mask pairs, covering 15 imaging modalities, over 30 cancer types, and a multitude of imaging protocols. The model aims to generalize SAM for medical segmentation in general, not exclusively for segmentation on H&E images, although it is also further trained on 6239 H&E images.

5. Our contribution

In line with our first research question (see section 2), our primary contribution aims to combine the efforts made in the works of MedSAM and CellViT. The former being an extension of SAM further trained on medical segmentation images, therefore possesses an encoder more familiar with medical images. We believe that the incorporation of medical images into the encoder, which have aspects more similar to H&E images than the "natural" images on which the classic SAM encoder is trained, can help improve the model's understanding of images and thus lead to better performance. This initial approach is illustrated in the figure 1.

Additionally, we tackle our second research question by

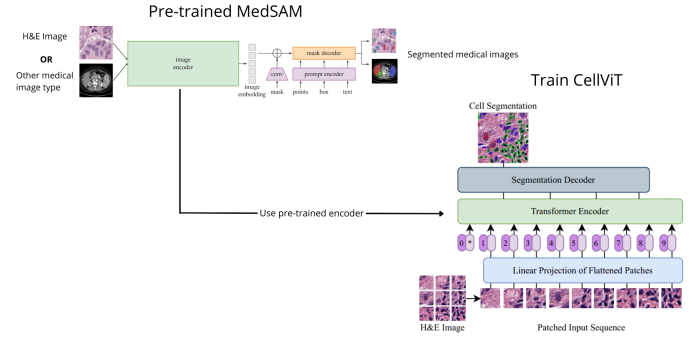


Figure 1. Instead of using the original weights of the SAM encoder, we use the weights of the MedSAM encoder, which is an extension of SAM that was further trained on medical images

evaluation the performances of SOTA model Hovernet with SAM used as a post-processing step. Our objective is to contrast the results obtained with this post-processing step with those presented in existing literature, particularly in terms of nuclei segmentation quality. Approach illustrated in figure 2

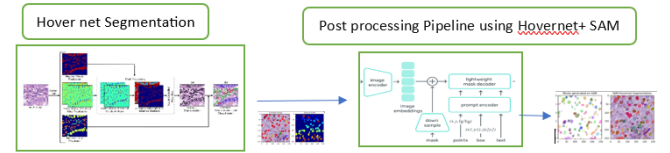


Figure 2. We use the information produced by HoverNet as input for SAM, which will allow for better segmentation

6. Hovernet with SAM as post-processing

6.1. Hovernet as a model

HoVer-Net model achieves SOTA performance as an interpret-able and reliable evaluation framework that effectively quantifies nuclear segmentation performance and overcomes the limitations of existing performance measures. This is achieved by performing nearest neighbor up-sampling via three distinct branches to simultaneously obtain accurate nuclear instance segmentation and classification. Name of the corresponding branches: (i) nuclear pixel (NP) branch; (ii) HoVer branch and (iii) nuclear classification (NC) branch. The NP branch predicts whether or not a pixel belongs to the nuclei or background, whereas the HoVer branch predicts the horizontal and vertical distances of nuclear pixels to their centers of mass.

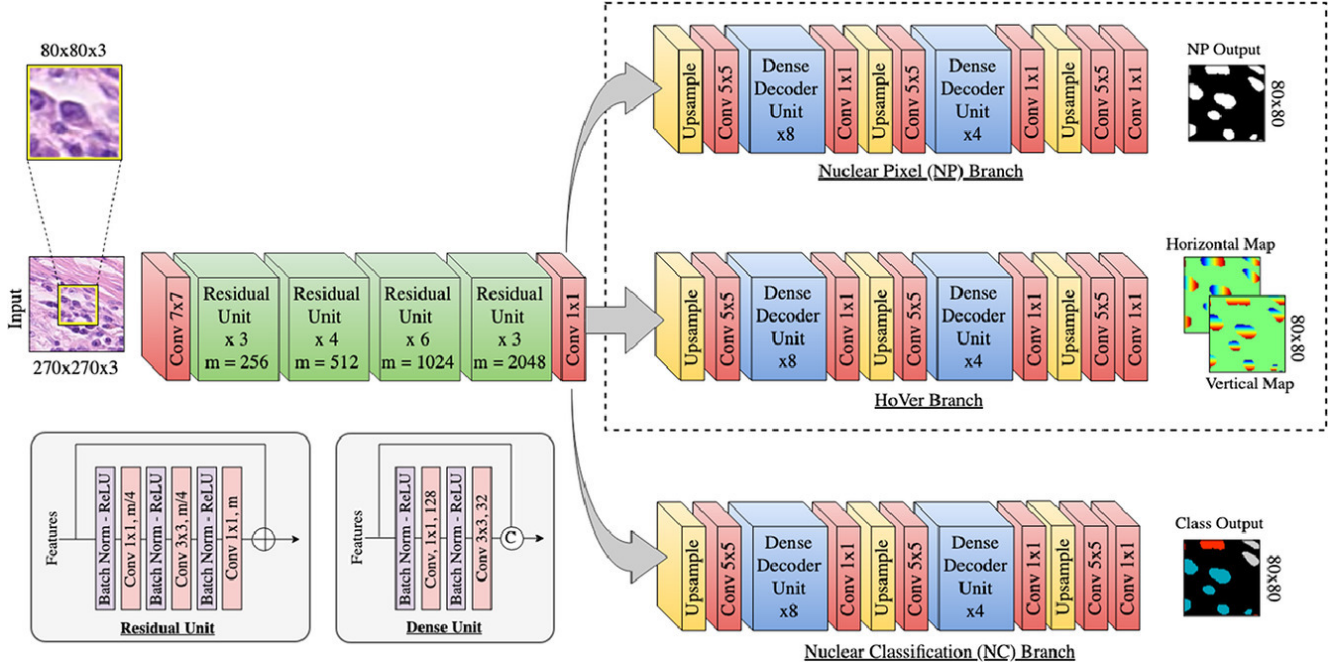


Figure 3. Hovernet Architecture

Then, the NC branch predicts the type of nucleus for each pixel. In particular, the NP and HoVer branches jointly achieve nuclear instance segmentation by first separating nuclear pixels from the background (NP branch) and then separating touching nuclei (HoVer branch). The NC branch determines the type of each nucleus by aggregating the pixel-level nuclear type predictions within each instance.

Overview of the architecture in figure 3 (a) (Pre-activated) residual unit, (b) dense unit. m indicates the number of feature maps within each residual unit. The yellow square within the input denotes the considered region at the output. When the classification labels are not available, only the up-sampling branches in the dashed box are considered. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article)

6.1.1 Segmentation and Classification

Figure 4, give the overview of the approach for simultaneous nuclear instance segmentation and classification. When no classification labels are available, the network produces the instance segmentation as shown in (a). The different colors of the nuclear boundaries represent different types of nuclei in (b).

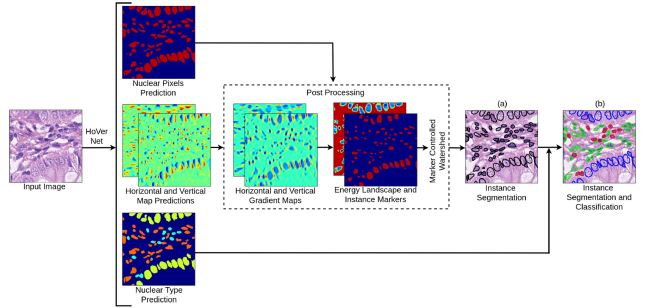


Figure 4

6.1.2 Horizontal and Vertical Maps

Within each horizontal and vertical map, pixels between separate instances have a significant difference. Therefore, calculating the gradient can inform where the nuclei should be separated. Since the output will give high values between neighbouring nuclei, where there is a significant difference in the pixel values. We define:

$$\text{Sobel}_m = \max(H_x(p_x), H_y(p_y)) \quad (1)$$

where p_x and p_y refer to the horizontal and vertical predictions at the output of the HoVer branch and H_x and H_y refer to the horizontal and vertical components of the Sobel operator. Specifically, H_x and H_y compute the horizontal and vertical derivative approximations and are shown by the gradient maps.

The highlights areas where there is a significant difference in neighbouring pixels within the horizontal and vertical maps. Therefore, areas such as the ones shown by the arrows in will result in high values within the Sobel operator. We compute markers

$$\mathbf{M} = \sigma(\tau((q, h)) - \tau((S_m, k))) \quad (2)$$

Here, $\tau(a, b)$ is a threshold function that acts on a and sets values above b to 1 or 0 otherwise. Specifically, h and k were chosen such that they gave the optimal nuclear segmentation results. σ is a rectifier that sets all negative values to 0 and q is the probability map output of the NP branch. We obtain the energy landscape.

$$E = [1 - \tau(S_m, k)] * \tau(q, h) \quad (3)$$

Finally, M is used as the marker during marker-controlled watersheds to determine how to split $\tau(q, h)$, given the energy landscape E . This sequence of events can be seen.

6.2. Results

6.2.1 Inference

Model weights obtained from training HoVer-Net can be supplied to process input images Tile/WSIs. When utilizing checkpoints trained on the PanNuke dataset, it's crucial to select the 'fast' model mode for inference, which employs a 256x256 patch input and 164x164 patch output. These checkpoints, originally trained using TensorFlow and converted to PyTorch format, are suitable for segmentation tasks and may include classification capabilities. It's essential to use the correct model mode especially for checkpoints trained exclusively for segmentation tasks.

6.2.2 Dice Score

At the output of NP and NC branches, we calculate the cross-entropy loss (Lc and Le) and the dice loss (Ld and Lf). These two losses are then added together to give the overall loss of each branch. Concretely, we define the cross entropy and dice losses as:

$$CE = -\frac{1}{n} \sum_{i=1}^N \sum_{k=1}^K X_{i,k}(I) \log Y_{i,k}(I) \quad (4)$$

$$Dice = 1 - \frac{2 \times \sum_{i=1}^N (Y_i(I) \times X_i(I)) + \epsilon}{\sum_{i=1}^N Y_i(I) + \sum_{i=1}^N X_i(I) + \epsilon} \quad (5)$$

In the current literature, DICE metrics have been mainly adopted to quantitatively measure the performance of nuclear instance segmentation. Given the ground truth X and

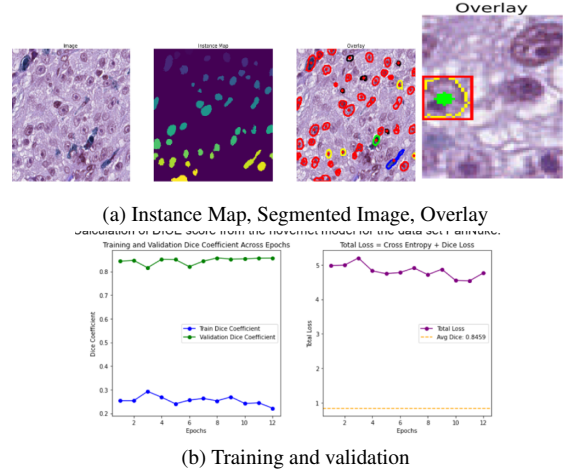


Figure 5. Overlay and Metric calculation

prediction Y , DICE2 computes and aggregates DICE per nucleus, where Dice coefficient (DICE) is defined as:

$$DICE = \frac{2 \times (X \cap Y)}{|X| + |Y|}$$

6.2.3 Post Processing Hovernet images Using SAM

The study explores the application of SAM to post-processed images generated by HoverNet. This investigation aims to assess if SAM can enhance the segmentation quality of HoverNet's outputs. Figure 6, SAM employs meta-learning techniques, enabling it to quickly adapt to various segmentation tasks with limited task-specific training data. By training on diverse segmentation tasks and leveraging meta-learning, SAM learns generalized features and segmentation strategies, allowing it to effectively segment objects or regions of interest in images.

Figure 6 depicts (a) Hovernet segmented image, (b) Ground truth of the image, (c) Masks generated by SAM for Hovernet image, (d) Final segmentation of hovernet image on SAM.

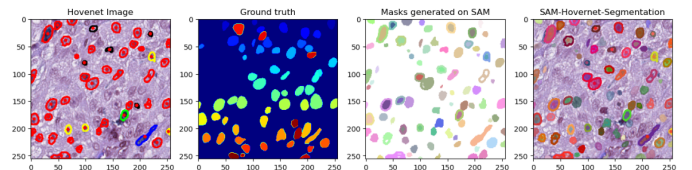


Figure 6. Post Processing with SAM

6.2.4 Metrics Evaluation

Further evaluation for the quality of segmented images was done by using metrics such as DICE, IOU, and Hausdorff distance. Comparing the Dice score of the HoverNet model to that of the SAM + HoverNet, the SAM + HoverNet demonstrates significantly higher DICE and IOU scores. Higher DICE scores indicate a greater degree of overlap between the predicted and ground truth segmentation masks, signifying improved accuracy and alignment in segmenting objects or regions of interest. This enhanced accuracy offers advantages such as better object delineation, more precise segmentation boundaries, and increased reliability in downstream tasks like image analysis, object detection, and medical diagnosis.

The discussion highlights the drawbacks of traditional metrics like DICE score and IOU for assessing segmentation models, particularly in the context of small structures such as nuclei. Instead, there's a recommendation to utilize instance-specific metrics that offer a more detailed evaluation of segmentation quality. Among these metrics, Hausdorff distance emerges as a notable choice, as it scrutinizes the dissimilarity between predicted and ground truth boundaries of individual nuclei, providing a finer assessment of segmentation accuracy.

Model	DICE	IOU	Hausdorff distance
Hovernet	0.84	-	544.68
SAM + Hovernet	0.95	0.94	267.361

Table 1. Comparison of DICE, IOU, and Hausdorff distance between HoverNet and SAM models.

Based on the comparison between the HoverNet and SAM + HoverNet models presented in the table, it is evident that the HoverNet + SAM model outperforms the HoverNet model in terms of several evaluation metrics. Specifically, the HoverNet + SAM model achieves a higher DICE score of 0.95 compared to 0.84 for the HoverNet model, indicating better overlap between predicted and ground truth nuclei instances. Additionally, in terms of Hausdorff distance, indicate how much the segmented regions deviate from the ground truth. A smaller distance suggests a better match between the segmented regions and the ground truth annotations, while a larger distance indicates more dissimilarity for single instance. Therefore, based on these metrics, the SAM + HoverNet model showcases better performance in segmenting nuclei instances compared to the HoverNet model.

7. CellViT with the encoder from MedSAM

7.1. CellViT as a model

CellViT's structure is particularly interesting for the tasks of segmenting and classifying cell nuclei within histopathological images such as those from the PanNuke dataset. It accomplishes this by combining the encoding power of a ViT with the decoding finesse inspired by HoVer-Net. In this part, we will specially focus on the encoding part as the hovernet-like decoder has the same architecture as the HoVerNet described above. Indeed, CellViT's encoder is the main part of its architecture, designed to interpret and process complex visual data from histopathological images. The choice of Vision Transformer is predicated on its superiority over traditional CNNs in current SOTA models. Unlike CNNs, which primarily capture local dependencies within images, Vision Transformers excel at understanding both the details and the broader context of the image. This capability is crucial for medical image analysis, where the distinction between different cell types and their states can be subtle yet significant.

Vision transformers segments the input image into a series of fixed-size patches. These patches are then flattened and transformed into a sequence of tokens, which are processed through multiple layers of the Transformer. This process allows the encoder to capture both local features of each patch and global relationships between patches across the entire image, enabling the model to leverage deep insights into the structure and distribution of cell nuclei. Now, in the era of foundational models, it's essential to leverage these enormous models to extract all their benefits.

In the case of CellViT, the authors tested several types of encoders but ultimately noted the best performance with the pre-trained SAM-B encoder (segment anything model) on 1.1 billion segmentation masks. This encoder is then connected to a HoVer-Net type decoder via skip connections. These connections are crucial for combining the detailed analysis provided by the encoder with the segmentation capabilities of the decoder. They ensure that the rich, contextual information captured by the encoder is not lost during decoding and is utilized to refine the segmentation output for a more accurate and nuanced understanding of the PanNuke images.

7.2. Losses

For faster training and better convergence of the network, a combination of different loss functions is employed for each network branch. The total loss function is :

$$L_{\text{total}} = L_{\text{NP}} + L_{\text{HV}} + L_{\text{NT}} + L_{\text{IC}} \quad (6)$$

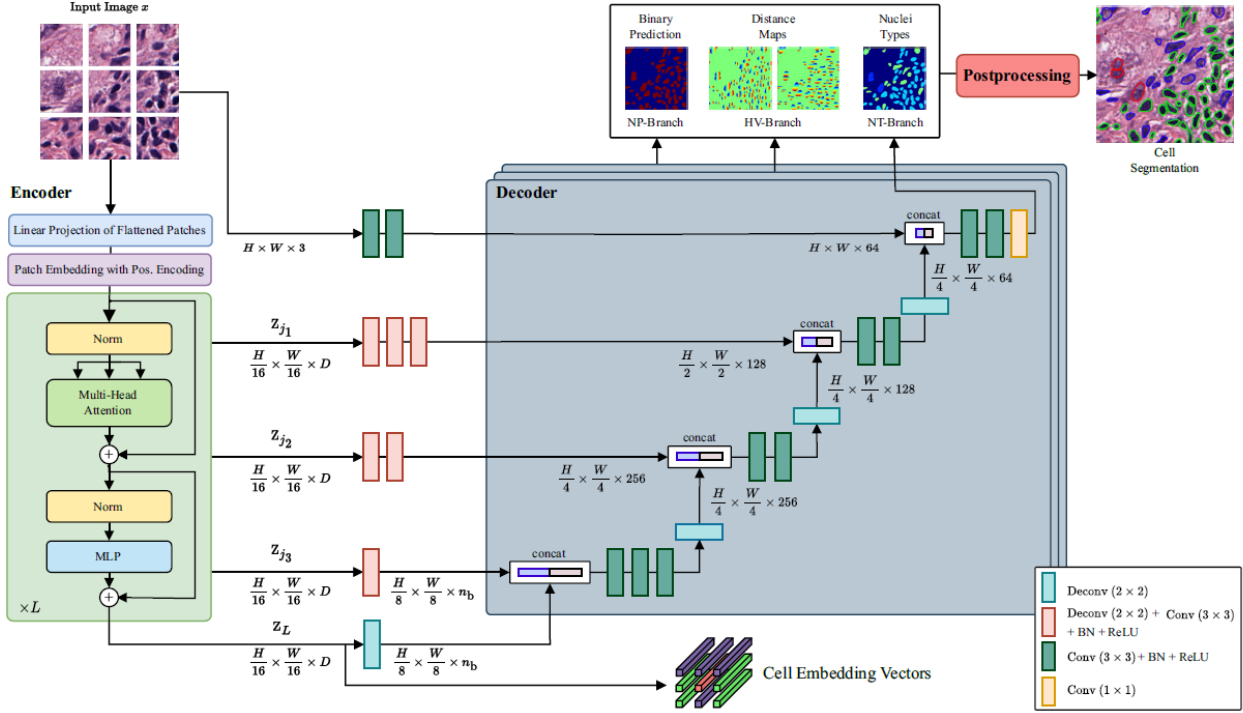


Figure 7. CellViT architecture

where L_{NP} denotes the loss for the NP-branch, L_{HV} the loss for the HV-branch, L_{NT} the loss for the NT-branch, and L_{TC} the loss for the TC-branch. Overall, the individual branch losses are composed of the following weighted loss functions:

$$L_{NP} = \lambda_{NP_{FT}} L_{FT} + \lambda_{NP_{DICE}} L_{DICE}$$

$$L_{HV} = \lambda_{HV_{MSE}} L_{MSE} + \lambda_{HV_{MSGE}} L_{MSGE}$$

$$L_{NT} = \lambda_{NT_{FT}} L_{FT} + \lambda_{NT_{DICE}} L_{DICE} + \lambda_{NT_{BCE}} L_{BCE}$$

$$L_{TC} = \lambda_{TC} L_{CE}$$

with the individual segmentation losses

$$L_{BCE} = -\frac{1}{n} \sum_{i=1}^{N_{px}} \sum_{c=1}^C y_{i,c} \log(\hat{y}_{i,c}) \quad (7)$$

$$L_{DICE} = 1 - \frac{2 \times \sum_{i=1}^{N_{px}} y_{i,c} \hat{y}_{i,c} + \varepsilon}{\sum_{i=1}^{N_{px}} y_{i,c} + \sum_{i=1}^{N_{px}} \hat{y}_{i,c} + \varepsilon} \quad (8)$$

and the cross-entropy as tissue classification loss :

$$L_{CE} = -\sum_{c=1}^{C_T} y_{cT} \log(\hat{y}_{cT}), \quad C_T = 19, \quad (9)$$

with the contribution of each branch loss to the total loss (6) controlled by the i -th hyperparameters λ_i . L_{MSE} denotes the mean squared error of the horizontal and vertical

distance maps and L_{MSGE} the mean squared error of the gradients of the horizontal and vertical distance maps, each summarized for both directions separately. In the segmentation losses (7)–(9), $y_{i,c}$ is the ground-truth and $\hat{y}_{i,c}$ the prediction probability of the i th pixel belonging to the class c , C the total number of nuclei classes, N_{px} the total amount of pixels, ε a smoothness factor and α_{FT} , β_{FT} and γ_{FT} are hyperparameters of the Focal Tversky loss L_{FT} . The Cross-Entropy loss (9) and Dice loss (8) are commonly used in semantic segmentation. To address the challenge of under-represented instance classes, the Focal Tversky loss, a generalization of the Tversky loss, is used. The Focal Tversky loss places greater emphasis on accurately classifying underrepresented instances by assigning higher weights to those samples. This weighting enhances the model's capacity to handle class imbalance and focuses its learning on the more challenging regions of the segmentation task.

7.3. Experiments

Moving on to our significant contribution to utilizing CellViT for our research project, we made the strategic decision to replace the CellViT encoder with a MedSam encoder. This encoder, fundamentally the same as the SAM encoder developed by Meta, has been retrained entirely on medical data. Our intuition behind this modification was grounded in the belief that integrating medical-

specific data on top of the more generic data used in Meta’s SAM model would yield better results. By tailoring the encoder to reflect the complexity and nuances of medical imagery more closely, we were expecting an improvement in model’s performance, particularly in the segmentation and classification of cell nuclei within histopathology images of PanNuke. As MedSAM encoder uses pretrained SAM-B weights, we have trained CellViT with SAM-B to stay coherent on our results.

7.4. Results

Initially, CellViT has been trained using one GPU Nvidia 1080 tx for 130 epochs where the encoder has been frozen for the first 25 epochs. To align with the original training, we used the exact same configuration as in the original paper with a learning rate scheduling with a scheduling factor of 0.85 to gradually reduce the learning rate during training. However, due to the limitation of computational resources, we have only trained for 40 epochs for 36 hours. However, we already have significant results with only 40 epochs.

Model	DICE	IOU
CellViT (SAM-B)	0.77	0.68
CellViT (MedSAM)	0.78	0.69

Table 2. Comparison of mean DICE and mean IOU between CellViT with SAM-B encoder and CellViT with MedSAM encoder after a training for 40 epochs

We can see in Table 2 and Table 3 that MedSAM allows slight improvements at the 40-th epoch and struggle to classify the same type of tissues such as Colon tissue. Those results might seem to be minor but in medical imaging this has a real impact as we are dealing with real people at the end. We are also confident that this Dice shift between both models is going to expand with further training.

8. Comparison of results (SAM as post-processing and as encoder)

Now, it is interesting to evaluate our experiments between themselves:

Looking at Table 4, we clearly see that SAM used for post processing offers the best advantages. Indeed, we were able to improve SOTA of CellViT using the method described in part 6. It would be interesting to see if we achieve the same performances with more computational power for further training of MedSAM.

Tissue Type	CellViT (SAM-B)	CellViT (Med-SAM)
Ovarian	0.8398	0.8362
Thyroid	0.7976	0.8219
Stomach	0.8546	0.8605
Uterus	0.8007	0.7818
Adrenal Gland	0.8009	0.8148
Bladder	0.75449	0.7411
Bile Duct	0.7607	0.7773
Liver	0.8427	0.861
Head and Neck	0.6586	0.6318
Pancreatic	0.8275	0.8457
Breast	0.8124	0.8114
Prostate	0.7973	0.8301
Testis	0.8083	0.8135
Colon	0.6833	0.7075
Esophagus	0.8192	0.8135
Cervix	0.6991	0.7625
Kidney	0.7417	0.8204
Skin	0.7201	0.7086
Lung	0.8019	0.8059

Table 3. Comparison of DICE Scores between CellViT (Med-SAM) and CellViT(SAM-B) Across Tissue Types

Model	DICE	IOU
SAM + HoverNet	0.95	0.94
CellViT (MedSAM)	0.78	0.69

Table 4. Comparison between our experiments : SAM as post processing vs SAM as encoder

9. Ethical and societal impact

We also believe it is important to emphasize that medical datasets, including PanNuke, typically do not indicate the origins of the patients who comprise the dataset. However, it has been shown that many datasets lack diversity [21]. Any model trained on such datasets thus incorporates their biases, and its performance on patients from ethnicity different from those predominantly represented in the dataset may prove to be inferior to those reported in the SOTA and in this paper. This is unacceptable, and there is an urgent need to create guaranteed diverse datasets in order to build models applicable to all patients.

Furthermore, we want to emphasize that integrating AI technologies into clinical practice requires adequate training and education for healthcare professionals. A proper understanding of how to interpret and utilize our pipelines is crucial to ensure safe and effective patient care while minimizing the risk of over reliance on automated systems. To this end, please refer to our GitHub repository: <https://github.com/medsam/medsam>

[//github.com/yanis92300/Lab_project.git](https://github.com/yanis92300/Lab_project.git)

10. Conclusion and perspectives

In conclusion, our exploration into the application of SAM in the context of cancer cell detection in HE images has shown promising results. By integrating SAM both as a post-processing step to Hovernet and as an encoder within the CellViT architecture, we have demonstrated its potential to enhance the segmentation quality and accuracy in identifying cancerous cells. The use of SAM as a post-processing tool significantly improved the segmentation quality, as evidenced by the substantial increase in the DICE score and reduction in Hausdorff distance, indicating a closer alignment of the segmented images with the ground truth.

On the other hand, incorporating the MedSAM encoder, an iteration of SAM encoder trained on medical images, into the CellViT model has shown slight improvements in segmentation performance. This adaptation suggests that pre-training on domain-specific data can provide marginal gains, showing the importance of relevant training datasets in model performance. However, the more substantial improvements seen with SAM's application as a post-processing step suggest that there remains significant room for exploration in leveraging SAM's capabilities more effectively.

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